

1. Write Regular Grammar for a^*
RLG: $S \rightarrow aS | \epsilon$ $w = aaa$ derivation:-
LLG: $S \rightarrow Sa | \epsilon$

2. Write RG for $(a+b)^*$

RLG: $S \rightarrow aS | bS | \epsilon$ LLG: $S \rightarrow Sa | Sb | \epsilon$

write RG for $(a+b)^+$

$S \rightarrow aS | bS | a | b$

$S \rightarrow Sa | Sb | a | b$

3. Write RG for $(ab)^*$

RLG: $S \rightarrow abS | \epsilon$ LLG: $S \rightarrow Sab | \epsilon$

4. Write RG over $\Sigma = \{a, b\}$ that will generate all strings that starts with a and end with b .

RE: $a(a+b)^*b$

RLG: $S \rightarrow aA$
 $A \rightarrow aA | bA | b$

LLG: $S \rightarrow Ab$
 $A \rightarrow Aa | Ab | a$

5. Write RG over $\Sigma = \{a, b\}$ that will generate all strings having substring aa .
RE = $(a+b)^*aa(a+b)^*$

RLG: $S \rightarrow aS | bS | aaA$
 $A \rightarrow aA | bA | \epsilon$

LLG: $S \rightarrow Sa | Sb | Aaa$
 $A \rightarrow Aa | Ab | \epsilon$

6. Write RG over $\Sigma = \{a, b\}$ that will generate all strings having different initial & final letter.

RE: $a(a+b)^*b + b(a+b)^*a$

RLG: $S \rightarrow aA | bB$
 $A \rightarrow aA | bA | b$
 $B \rightarrow aB | bB | a$

LLG: $S \rightarrow Ab | Ba$
 $A \rightarrow Aa | Ab | a$
 $B \rightarrow Ba | Bb | b$

7. Design RG for $(ab^*c)^*$

RLG: $S \rightarrow aA | \epsilon$
 $A \rightarrow bA | cS$

LLG: $S \rightarrow Ac | \epsilon$
 $A \rightarrow Ab | aSa$

8. Write RG over $\Sigma = \{a, b\}$ such that string contain atmost two a's.

RE: $b^* + b^*ab^* + b^*ab^*ab^*$

RLG: $S \rightarrow bS | \epsilon | aA$
 $A \rightarrow bA | \epsilon | aB$
 $B \rightarrow bB | \epsilon$

LLG: $S \rightarrow Sb | \epsilon | AA$
 $A \rightarrow Ab | \epsilon | Ba$
 $B \rightarrow Bb | \epsilon$

2. Equivalence b/w RG & FA! - $\emptyset$

Q. Conversion from FA to RLG! - (From FA to RG)

I/P! - FA (M) output! RLG such that $L(G) = L(M)$

(i) Variables corresponds to states

(ii) Terminals corresponds to input alphabet

(iii) Start symbol corresponds to initial states

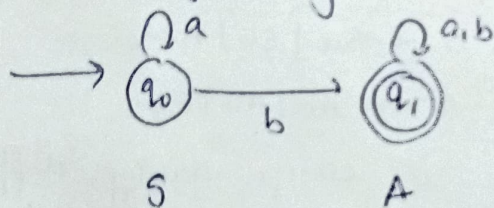
(iv) P (Set of production rules) is defined by following rules!

a. $A_i \rightarrow a A_j$ is included in P if $\delta(q_i, a) = q_j \notin F$

b. $A_i \rightarrow a A_j$ and $A_i \rightarrow a$ are included in P if $\delta(q_i, a) = q_j \in F$

* $S \rightarrow \epsilon$ is included only if initial state is also one of the final state.

Q. Construct RG generating the regular set representing by $P = a^*b(a+b)^*$

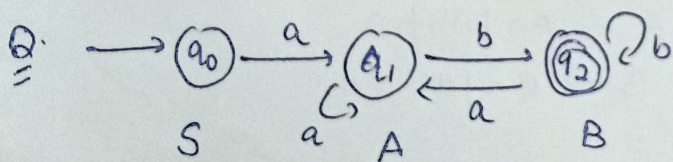


$\Sigma = \{a, b\}$

$V = \{S, A\}$

$\delta(q_0, a) = q_0$	$S \rightarrow aA$
$\delta(q_0, b) = q_1$	$S \rightarrow bA$
	$S \rightarrow b$
$\delta(q_1, a) = q_1$	$A \rightarrow aA$
	$A \rightarrow a$
$\delta(q_1, b) = q_1$	$A \rightarrow bA$
	$A \rightarrow b$

$P \Rightarrow S \rightarrow aS \mid bA \mid b$
 $A \rightarrow aA \mid bA \mid a \mid b$

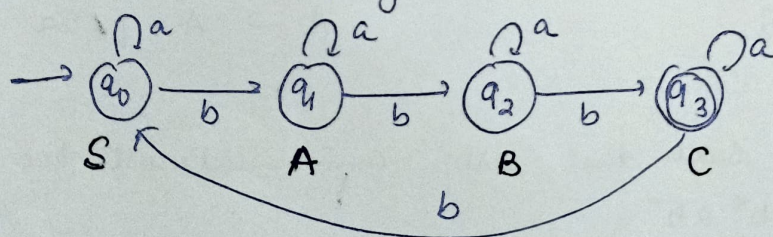


$\Sigma = \{a, b\}$

$V = \{S, A\}$

$S \rightarrow aA$	$B \rightarrow bB \mid b$
$A \rightarrow aA$	$B \rightarrow aA$
$A \rightarrow bB \mid b$	

Q. Construct a RG accepting $L = \{w \mid w \in \{a, b\}^* \text{ and no. of b's in } w \text{ is } 3 \pmod{4}\}$



$\Sigma = \{a, b\}$

$V = \{S, A, B, C\}$

$P \Rightarrow S \rightarrow aS$
 $S \rightarrow bA$
 $A \rightarrow aA \mid bB$
 $B \rightarrow aB \mid bC \mid b$
 $C \rightarrow aC \mid a \mid bS$

b. Conversion from RLG to FA: (from RLG to FA)

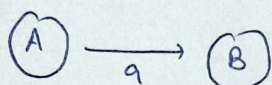
≡ P: Given RLG (G) O/p FA (M) such that $L(M) = L(G)$

* Number of states in automata will be more ^{or equal} than number of non-terminal (generally one more)

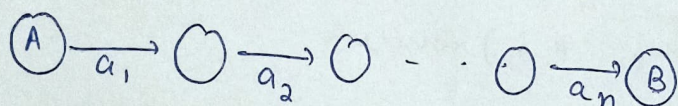
* Initial state corresponds to start symbol.

* The transition of automata is obtained as follows:

i. For every production $A \rightarrow aB$, include $\delta(A, a) = B$ that is make a transition from A to B on labeled 'a'.

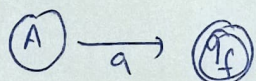


Similarly, for production $A \rightarrow a_1 a_2 \dots a_n B$,

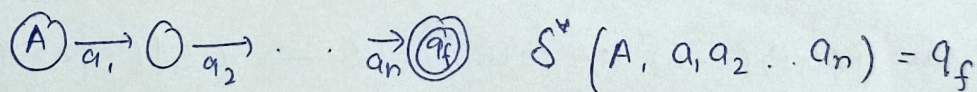


$$\delta^*(A, a_1 a_2 \dots a_n) = B$$

ii) For production $A \rightarrow a$, then make transition $\delta(A, a) = q_f$



Similarly, for production $A \rightarrow a_1 a_2 \dots a_n$

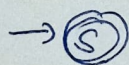


iii) For every production $A \rightarrow \epsilon$, make $\delta(A, \epsilon) = A$ & A will be final state.

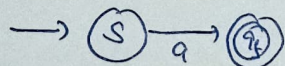
Ex: RLG

FA

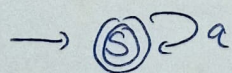
$S \rightarrow \epsilon$



$S \rightarrow a$



$S \rightarrow aS / \epsilon$

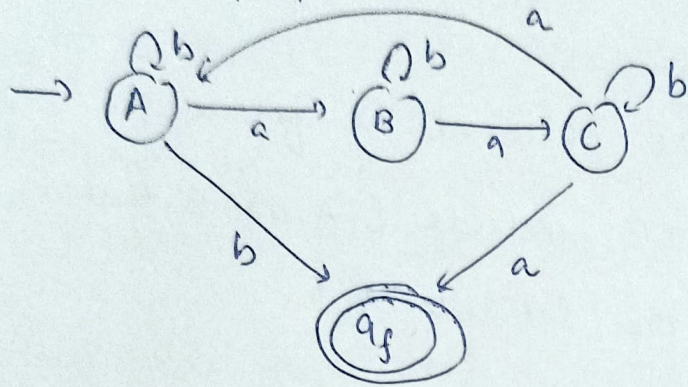


Q. Construct a Finite Automata from the given RLG:-

$$A \rightarrow aB / bA / b$$

$$B \rightarrow aC / bB$$

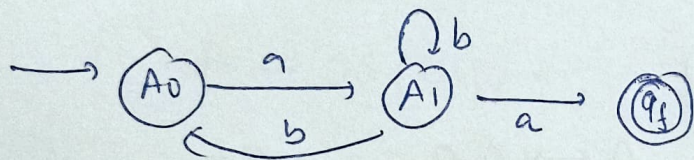
$$C \rightarrow aA / bc / a$$



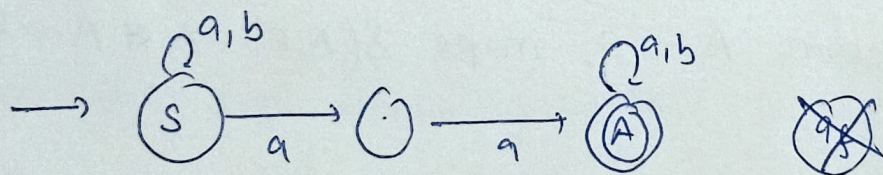
Q. Construct a Finite Automata for given grammar:

$$G = (\{A_0, A_1\}, \{a, b\}, P, A_0) \text{ where}$$

$$P = \{ A_0 \rightarrow aA_1, A_1 \rightarrow bA_1, A_1 \rightarrow a, A_1 \rightarrow bA_0 \}$$



Q. Given RLG $\Rightarrow (a+b)^* aa (a+b)^*$ $S \rightarrow aS / bS / aaA$
 $A \rightarrow aA / bA / \epsilon$



Q: Conversion of LLG from DF FA:-

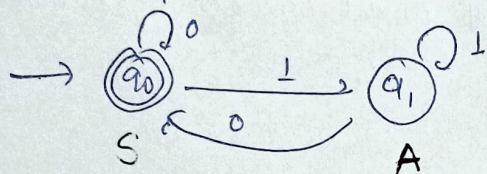
Step 1: Interchange the positions of initial & final states.

Step 2: Reverse the directions of all the transitions, keeping all the intermediate steps unchanged.

Step 3: Write LL production from new transition graphs.

* In case of more final states, create a new FS and add ϵ transitions to all old FS.

Q: Find the equivalent RLG & LLG for the following FA



$\Sigma = \{0, 1\}$

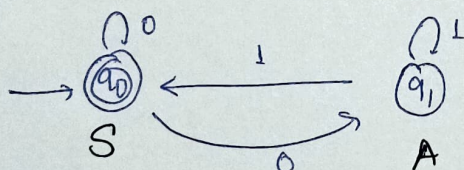
$V = \{S, A\}$

RLG: $P = S \rightarrow OS \mid 0 \quad S \rightarrow \epsilon$

$S \rightarrow 1A$

$A \rightarrow OS \mid 0 \mid 1A$

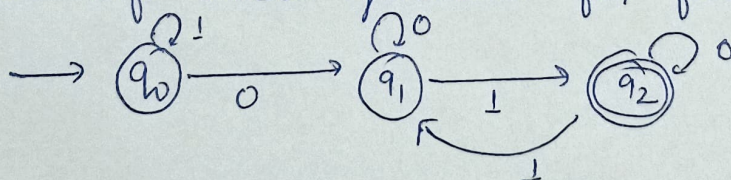
LLG:



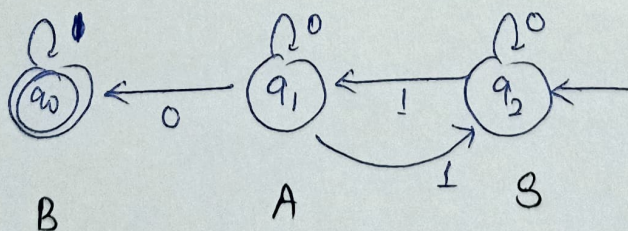
$P \Rightarrow S \rightarrow SO \mid A0 \mid 0$

$A \rightarrow A1 \mid S1 \mid 1$

Q: Find the left linear grammar for following DFA



Reverse:-



$\Sigma = \{0, 1\}$

$V = \{S, A, B\}$

$P \Rightarrow S \rightarrow SO \mid A1$

$A \rightarrow A0 \mid B0 \mid 0 \mid S1$

$B \rightarrow B1 \mid 1$